

- 4) Eight adjacent squares represent a term of one literal.
- 5) Sixteen adjacent squares represent the function equal to 1.

Example 3-5: Simplify the Boolean function;

$$F(w, x, y, z) = \Sigma(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$$

wz	00	01	11	10
00	1	1		1
01	1	1		1
11	1	1		1
10	1	1		

• Eight adjacent squares marked with 1's can be combined to form the one literal term y' .

• The remaining three 1's on the right cannot be combined together to give a simplified term. They must be combined as four adjacent squares.

Figure 3.9 Map for Example 3-5

Since the larger the number of squares combined, the less the number of literals in the term.

• In this example, the top two 1's on the right are combined with the top two 1's on the left to give the term $w'z'$.

• The squares comprising the two middle rows and the two end columns, giving the term xz' . The simplified function is:

$$F = y' + w'z' + xz'$$

Example 3-6: Simplify the Boolean function:

$$F = A'B'C' + B'CD' + A'BCD' + AB'C'$$

AB	00	01	11	10
00	1	1		1
01				1
11				
10	1	1		1

This function has four variables and, as expressed, consists of three terms, each with three literals, and one term of four literals. Each term of three literals is represented in the map by two squares. For

example, $A'B'C'$ is represented in squares 0000 and 0001. The function can be simplified in the map by taking the 1's in the four corners to give the term $B'D'$. The two left-hand 1's in the top row are combined with the two 1's in the bottom row to give the term $B'C'$.